

Functional Analysis — Homework 1

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Problem 1. For $p \in [1, \infty)$, give an example of a sequence which is in ℓ^q for all $q > p$, but not in ℓ^p .

Proof. Consider the sequence (x_n) defined by

$$x_n = \left(\frac{1}{n}\right)^{\frac{1}{p}}.$$

Then $\|(x_n)\|_p^p = \sum_1^\infty 1/n$, which is the harmonic series, so $(x_n) \notin \ell^p$, but for any $q > p$ we have $(1/n)^{q/p} < 1/n$ when $n \neq 1$. Thus $\|(x_n)\|_q^q$ converges and $(x_n) \in \ell^q$. $\square$

Problem 2. If a subspace S of a normed space V contains a nonempty open set, then $S = V$.

Proof. Let U be a nonempty open subset of S , let $p \in U$, and let $\delta > 0$ such that $B_\delta(p) \subset U$. Now consider an arbitrary $x \in V$ and $\alpha > 0$. The idea is to “nudge p in the direction of x ” while remaining in $B_\delta(p)$ by finding α such that $\alpha(x - p) + p$ is close enough to p . Indeed,

$$\|[\alpha(x - p) + p] - p\| = \|\alpha(x - p)\| = \alpha\|x - p\|,$$

so by choosing $\alpha < \delta/\|x - p\|$, we get $\alpha(x - p) + p \in B_\delta(p) \subset S$. We can now use the closure of S under addition and scalar multiplication to get that

$$[\alpha(x - p) + p] + [\alpha p - p] = \alpha x \in S,$$

and thus multiplying by $1/\alpha$ we get $x \in S$. Since x was arbitrary, we have shown that $V \subset S$, and thus $V = S$. $\square$

Problem 3. Show that in any normed space V , the operations

$$+ : V \times V \rightarrow V \quad \cdot : \mathbb{R} \times V \rightarrow V$$

of addition and scalar multiplication are continuous.

Proof. NOTE: I haven’t been able to complete either part.

First we do scalar multiplication. Let $(x_n) \rightarrow x$ in V and $(c_n) \rightarrow c$ in $\mathbb{R}$. Then given $\varepsilon > 0$, we can find $N \in \mathbb{N}$ such that when $n > N$, $\|x - x_n\| < \sqrt{\varepsilon}$ and $|c - c_n| < \sqrt{\varepsilon}$. Then

$$\begin{aligned} \varepsilon &> \|(c - c_n)(x - x_n)\| \\ &= \|cx - c_nx - cx_n + c_nx_n\|. \end{aligned}$$

We could imagine trying to throw away the $c_n x$ and $c_n x_n$ terms to get an inequality, but the more important difficulty is that we need constraints on $\|cx - c_n x_n\|$ to prove convergence of $(c_n x_n)$, and we only have $\|cx + c_n x_n\|$.

For addition, let $(x_n) \rightarrow x$ and $(y_n) \rightarrow y$ in V . Then given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ (take such an integer for (x_n) and one for (y_n) and then N is the max of these two) such that when $n > N$,

$$\begin{aligned}\frac{\varepsilon}{2} &> \|x - x_n\| \geq \|x\| - \|x_n\| \\ \frac{\varepsilon}{2} &> \|y - y_n\| \geq \|y\| - \|y_n\|.\end{aligned}$$

We can then add these together to get

$$\begin{aligned}\varepsilon &\geq \|x\| + \|y\| - (\|x_n\| + \|y_n\|) \\ &\geq \|x + y\| - (\|x_n\| + \|y_n\|),\end{aligned}$$

but the issue here is that we can't pull the same triangle inequality trick (nor any other trick I can think of) to finish it off and get a

$$\geq \|x + y\| - \|x_n + y_n\|$$

line, so I'm a bit stuck. □

Problem 4. Let V be a normed space. Show that a linear map $f : V \rightarrow \mathbb{R}$ is continuous if and only if $\ker(f)$ is closed.

Proof. If f is continuous, then since $\ker(f)$ is the preimage of the closed set $\{0\}$, $\ker(f)$ is closed. Now if $\ker(f)$ is closed, let (x_n) be a sequence converging to x in V . We want to show that $(f(x_n)) \rightarrow f(x)$, but I haven't been able to make any progress towards that. The only thing that seems a bit promising is "translating" the sequence so that it becomes $(x_n - x) \rightarrow 0$. This gets me a zero to potentially work with... but I haven't figured out how to make use of it. □

Problem 5. Let $\rho : V \rightarrow [0, +\infty)$ be a seminorm. Prove that if there exists a positive number $\delta > 0$ such that the preimage $\rho^{-1}[0, \delta]$ contains an open ball, then ρ is continuous.

Proof. I'm super stuck on this one. Interested to see the solution and think through it, but even after talking with you I'm having trouble really formulating the argument. There's something about scaling it and trapping a sequence within the open ball after maybe translation... but I haven't been able to make any real headway. □

Problem 6. Let V be a normed space. Show that V is Banach if and only if every absolutely convergent series is convergent.

Proof. First, assume V is Banach and $\sum \|x_i\| < \infty$. Then $(\|x_i\|)$ is Cauchy in $\mathbb{R}$, so given $\varepsilon > 0$, we can choose $N > 0$ such that when $m > n > N$,

$$\sum_1^m \|x_i\| - \sum_1^n \|x_i\| = \sum_{n+1}^m \|x_i\| < \varepsilon.$$

Then we also have that

$$\left\| \sum_1^m x_i - \sum_1^n x_i \right\| = \left\| \sum_{n+1}^m x_i \right\| \leq \sum_{n+1}^m \|x_i\| < \varepsilon,$$

so the partial sums of $\sum_1^\infty x_i$ are also Cauchy and thus converge since V is Banach.

Now if every absolutely convergent series is convergent, then let (x_i) be a Cauchy sequence in V . Define $y_1 := x_1$, and then recursively define $y_n := x_n - \sum_{i=1}^{n-1} y_i$ for all $n > 1$, so that $x_n = \sum_1^n y_i$ for all n . We then just need to show that $\sum y_i$ is absolutely convergent, and we will have shown that x_n converges and thus V is Banach. . .

But I've been unable to figure out how to do that. □

Problem 7. In this problem, you will show that for any number $p > 1$, the function

$$\|x\|_p = \|(x_1, \dots, x_n)\|_p = (|x_1|^p + \dots + |x_n|^p)^{1/p}$$

is a norm on $\mathbb{R}^n$. The extension to ℓ^p follows easily from that. It will be helpful to define a number q , called the conjugate exponent to p , by the equation: $\frac{1}{p} + \frac{1}{q} = 1$.

- (a) (Young's inequality) Use the fact that $\ln(x)$ is a concave function to prove that if a, b are positive numbers, then

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

- (b) (Hölder's inequality) Given any $x, y \in \mathbb{R}^n$, use Young's inequality to prove that

$$|\langle x, y \rangle| \leq \|x\|_p \|y\|_q$$

where $\langle x, y \rangle$ denotes the standard dot product in $\mathbb{R}^n$.

- (c) (Minkowski's inequality) Given any $x, y \in \mathbb{R}^n$, use Hölder's inequality to prove

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p,$$

which is the triangle inequality for $\|\cdot\|_p$.

Proof.

- (a) Since $\ln(x)$ is concave and the tangent line at x_0 is

$$\begin{aligned} y &= \frac{1}{x_0}(x - x_0) + \ln(x_0) \\ &= \frac{x}{x_0} - 1 + \ln(x_0) \dots \end{aligned}$$

We can rewrite the statement as

$$\begin{aligned} ab &\leq \frac{a^p}{p} + \frac{b^p}{p} \\ &= \frac{a^p}{p} + \frac{b^{\frac{p}{p-1}}}{\frac{p}{p-1}} \\ &= \frac{a^p + (p-1)b^{\frac{p}{p-1}}}{p} \dots \end{aligned}$$

not sure where any of that gets me.

(b) We have

$$\begin{aligned} |\langle x, y \rangle| &= \left| \sum_1^n x_i y_i \right| \\ &\leq \sum_1^n |x_i| |y_i| \\ &\leq \sum_1^n \frac{|x_i|^p}{p} + \frac{|y_i|^q}{q} \\ &= \sum_1^n \frac{|x_i|^p}{p} + \sum_1^n \frac{|y_i|^q}{q} \\ &= \frac{1}{p} \sum_1^n |x_i|^p + \frac{1}{q} \sum_1^n |y_i|^q \\ &= \frac{\|x\|_p^p}{p} + \frac{\|y\|_q^q}{q}, \end{aligned}$$

which feels like we're close to a solution, but in fact Young's Inequality tells us that

$$\|x\|_p \|y\|_q \leq \frac{\|x\|_p^p}{p} + \frac{\|y\|_q^q}{q},$$

so our first attempt is not tight enough, and I'm not sure where to go from there.

(c) I really have no idea how to start with this one. The only idea I was able to come up with was that if we could somehow end up with a $\|x\|_p^{1/p} \|y\|_q^{1/q}$, we would get

$$\|x\|_p^{1/p} \|y\|_q^{1/q} \leq \frac{\|x\|_p}{p} + \frac{\|y\|_q}{q},$$

and you could play around with some scalars there, but I'm not sure what gets you from using both p and q to just using p . I'm also not sure how you would get to that very contrived scenario anyways.

□